

Disorder in Cavity-Coupled Jahn–Teller Active Molecules

From Single-Molecule Polaritons to Disorder-Averaged Collective Spectra

Ann Lia Sam

NISER, Bhubaneswar

Under the guidance of
Prof. Krishna R. Nandipati and Prof. Athreya Sankar, IIT Madras

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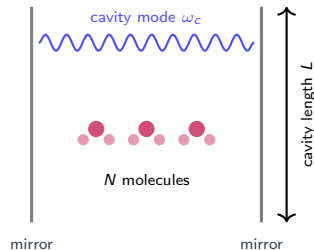
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Outline

- 1 Cavity–matter interaction
- 2 Jahn–Teller active molecules
- 3 Single molecule: cavity Jahn–Teller polaritons (2023)
- 4 Collective vibronic cascade (2025)
- 5 Introducing disorder
- 6 Disorder in our model & results

Molecules in an optical cavity: polaritons

- A Fabry-Pérot cavity confines the electromagnetic field into discrete **standing-wave modes** of frequency ω_c set by the mirror spacing L .
- When the light-matter coupling exceeds all loss rates (*strong coupling*), molecule and photon form hybrid eigenstates: **polaritons**.⁶
- These split into a **lower** (LP) and **upper** (UP) polariton, separated by the **Rabi splitting** Ω .
- Coupling N molecules collectively enhances the interaction as $\sqrt{N}$: the single-molecule coupling is $g = \Omega/(2\sqrt{N})$, where $N=1$.



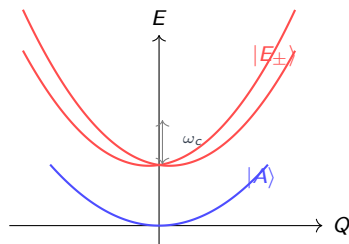
Collective strong coupling:
LP/UP split by $\Omega = 2\sqrt{N}g$.

The Jahn–Teller effect and the $(E \times e)$ model

Jahn-Teller theorem.⁵ A non-linear molecule in an orbitally degenerate electronic state spontaneously distorts to lift the degeneracy.

$(E \times e)$ problem. A doubly-degenerate electronic state E (components $|E_x\rangle, |E_y\rangle$) coupled to a doubly-degenerate vibrational mode e (coordinates Q_x, Q_y):

- ground state $|A\rangle$, excited doublet $|E_{\pm}\rangle = (|E_x\rangle \pm i|E_y\rangle)/\sqrt{2}$;
- potential surfaces form a *Mexican hat* with a **conical intersection**;



Excited $(E \times e)$ surfaces (red) and ground $|A\rangle$ (blue) vs. mode coordinate Q .

Cavity Jahn–Teller Hamiltonian — one molecule

There are circular modes $\hat{a}_{\pm}$ carrying photon angular momentum $\pm\hbar$.¹ The cavity–JT Hamiltonian is $\hat{H} = \hat{H}_{\text{JT}} + \hat{H}_C + \hat{H}_{CM}$, with

$$\hat{H}_{\text{JT}} = \sum_{r=+,-} [\hbar\omega \hat{b}_r^\dagger \hat{b}_r + \epsilon |E_r\rangle\langle E_r|] + \kappa [(\hat{b}_+^\dagger + \hat{b}_-) |E_-\rangle\langle E_+| + \text{h.c.}],$$

$$\hat{H}_C = \hbar\omega_c (\hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-), \quad \hat{H}_{CM} = \frac{\Omega}{2} (\hat{a}_+^\dagger |A\rangle\langle E_+| + \hat{a}_-^\dagger |A\rangle\langle E_-| + \text{h.c.}).$$

Operators. $\hat{b}_{\pm} = (\hat{b}_x \mp i\hat{b}_y)/\sqrt{2}$ - ladder operators of the degenerate e vibration carrying $\pm\hbar$ vibrational angular momentum; $\hat{a}_{\pm}$ - annihilation and creation operators of the two circularly-polarized ($\pm\hbar$) cavity photons.

The JT (vibronic) coupling κ exchanges angular momentum *between the electronic and vibrational subsystems of the molecule*; the cavity coupling $\hat{H}_{CM}$ then exchanges it with the photon.

Angular momenta & conservation

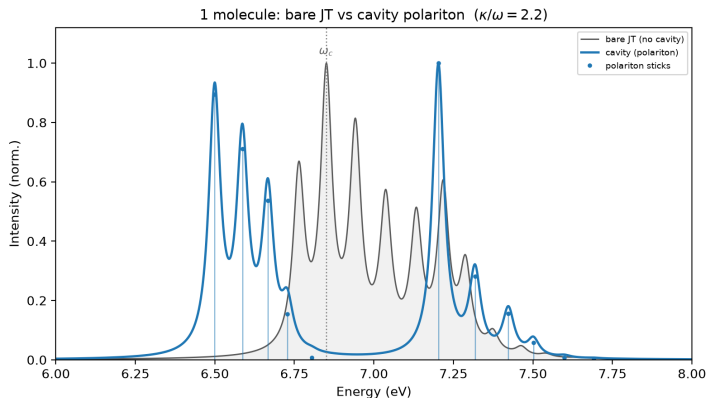
$$\hat{L}_z = \hbar(\hat{b}_+^\dagger \hat{b}_+ - \hat{b}_-^\dagger \hat{b}_-) \quad \text{vibrational}$$

$$\hat{S}_z = \hbar(|E_+\rangle\langle E_+| - |E_-\rangle\langle E_-|) \quad \text{electronic}$$

$$\hat{l}_z = \hbar(\hat{a}_+^\dagger \hat{a}_+ - \hat{a}_-^\dagger \hat{a}_-) \quad \text{photonic}$$

$$\boxed{\hat{J} = 2\hat{L}_z + \hat{S}_z + \hat{l}_z} \text{ is conserved.}$$

Single-molecule polaritonic spectrum (reproduced)



- Cavity ($\omega_c = 6.85$ eV) resonant with the brightest bare JT transition; the bare JT spectrum (grey) is reshaped into polaritons.
- Splits into **LP** and **UP** branches of **polaritonic peaks** - each a mixed light-matter (vibronic + photon) state, shown as sticks.
- For $N = 1$ only **3 vibronic sectors** $v \in \{0, -1, -2\}$ are accessible (bounded).

Parameters: $\epsilon = 7$ eV, $\omega = 0.082$ eV,
 $\kappa/\omega = 2.2$, $\Omega/2\omega_c = 0.05$; $g = \Omega/2$.¹

Collective Cavity Coupling (N molecules)

N identical JT molecules² coupled to the two cavity modes ($\hat{H}_{m,k}$ is the single-molecule JT Hamiltonian of the previous slide):

$$\hat{H} = \hat{H}_c + \sum_{k=1}^N (\hat{H}_{m,k} + \hat{H}_{m-c,k}), \quad \hat{H}_c = \hbar\omega_c (\hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-),$$

$$\hat{H}_{m-c,k} = \frac{\Omega}{2\sqrt{N}} (\hat{a}_+^\dagger |A\rangle \langle E_+ |_k + \hat{a}_-^\dagger |A\rangle \langle E_- |_k + \text{h.c.}).$$

Two conserved quantities:

Excitation number

$$\hat{N}_{\text{ex}} = \sum_k (|E_+\rangle \langle E_+ |_k + |E_-\rangle \langle E_- |_k) + \hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-$$

Total angular momentum

$$\hat{J} = \sum_{k=1}^N \hat{V}_k + \hat{P}, \quad \hat{V}_k = 2\hat{L}_{z,k} + \hat{S}_{z,k}$$

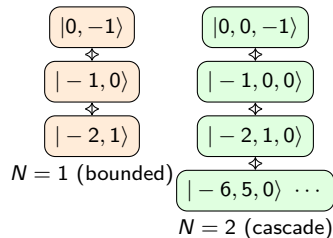
We work in the optically-relevant single-excitation sector ($n_{\text{ex}} = 1$, $j = -1$).

The Collective Vibronic cascade

Each molecule's JT Hamiltonian has vibronic eigenstates with a definite **vibronic angular momentum** $\nu = 2(n_+ - n_-) + S_z$, where n_+, n_- are the vibrational quanta (Fock occupation numbers) of the b_+, b_- modes:

$$\hat{H}_{m,k} = \sum_{\nu} \sum_i \lambda_i^{\nu} |\lambda_i^{\nu}\rangle \langle \lambda_i^{\nu}|_k$$

- $N = 1$: bounded - starting from the ground state only $\nu \in \{0, -1, -2\}$ are reachable.
- $N \geq 2$: the cavity photon mediates the transfer of angular momentum between molecules $\Rightarrow$ an unbounded ladder of ν sectors - the **collective vibronic cascade**.



The basis-state chains above show the sectors populated starting from the ground state + one RCP photon.

Participation Ratio and the $P(v)$ Heatmap

To *quantify* how far the cascade spreads, we decompose each polaritonic eigenstate $|\psi\rangle = \sum_{a,b,p} C_{a,b,p} |a, b, p\rangle$ over single-molecule vibronic sectors v :

Sector Population $P(v)$

$$P^{(k)}(v_0) = \sum_{a: v_a=v_0} \sum_{b,p} |C_{a,b,p}|^2, \quad \sum_v P(v) = 1.$$

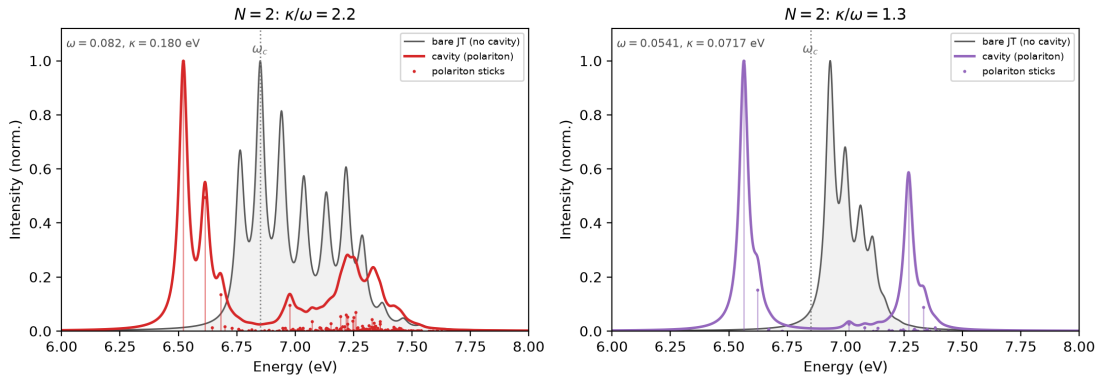
Participation Ratio

$$\text{PR} = \frac{1}{\sum_v P(v)^2}$$

- PR counts how many v sectors of an eigenstate participate meaningfully: $\text{PR} = 1$ if localized in one sector, larger if spread over many.
- $N = 1$: $\text{PR} \leq 3$ (bounded). $N = 2$: PR up to ~ 20 .
- The $P(v)$ **heatmap** plots, for every bright polaritonic state (x , energy-ordered), its $P(v)$ over sectors v (y), colored by value — the paper's Fig. S1.²

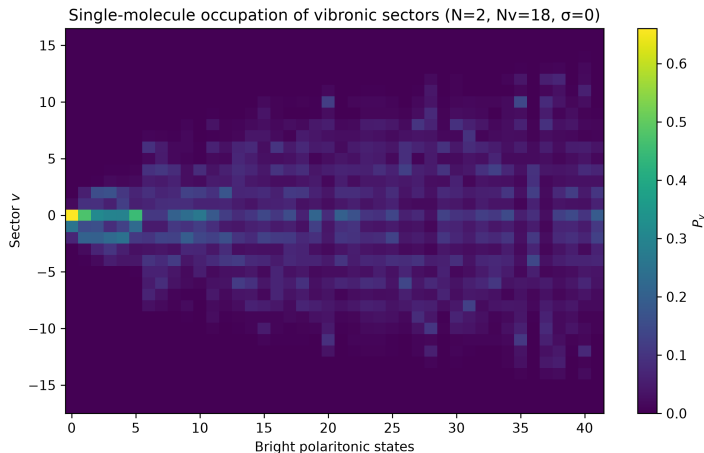
Two-Molecule Polaritonic Spectra (Reproduced)

Two-molecule cavity JT spectra (reproduced), $\omega_c = 6.85$ eV, $\Omega/2\omega_c = 0.05$



Bare JT (grey) vs. two-molecule cavity polariton spectrum (colored) with the polaritonic **stick spectrum**, for two vibronic-coupling strengths. Left: $\kappa/\omega = 2.2$ ($\omega = 0.082$, $\kappa = 0.180$ eV). Right: $\kappa/\omega = 1.3$ ($\omega = 0.0541$, $\kappa = 0.0717$ eV). Both at $\omega_c = 6.85$ eV, $\Omega/2\omega_c = 0.05$, $N = 2$.

Vibronic-Sector Heatmap $P(v)$ — clean $N = 2$ (reproduced)



For every *bright* polaritonic eigenstate we plot the probability $P(v)$ that a molecule occupies vibronic sector v (definition on the previous slide).

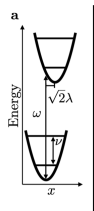
- x-axis: bright states (energy-ordered); y-axis: sector v ; color: $P(v)$.
- $N = 2$ ($\#$ vibrational states, $N_v = 18$): a broad, well-converged cascade with $PR_{\max} \approx 20$.

Why disorder?

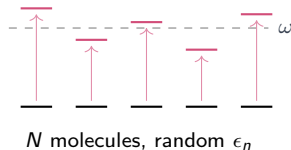
Real molecular ensembles are **inhomogeneous**: each molecule has a slightly different transition energy (static *energy disorder*). This work frames what disorder does to a cavity-coupled ensemble of two-level molecules:

Disordered Holstein–Tavis–Cummings (with vibrations)⁴

Each molecule is a two-level system on a displaced harmonic potential (Holstein coupling). Central result: **disorder enhances** excitation transfer and vibrational entanglement, and dark states *acquire photon weight* (brighten). Effect peaks when disorder $W \sim g_c$.



Two-level molecule: displaced harmonic surfaces, spacing ν .



Many two-level molecules with disordered transition energies $\omega + \epsilon_n$.

Disorder: The Equations

Disordered Holstein–Tavis–Cummings model⁴:

$$\hat{H} = \hat{H}_{\text{TC}} + \hat{H}_{\text{vib}} + \hat{H}_{\text{H}} + \hat{H}_{\text{dis}},$$

$$\hat{H}_{\text{dis}} = \sum_n \epsilon_n \hat{\sigma}_n^+ \hat{\sigma}_n^-$$

with energy offsets ϵ_n of mean 0 and variance W^2 (Gaussian), and

$$\hat{\sigma}_n^+ = |E_+\rangle\langle E_+|_n, \quad \hat{\sigma}_n^- = |E_-\rangle\langle E_-|_n$$

the **projection operators onto the E_+ / E_- excited sublevels of the n -th molecule** (same $E_{\pm}$ labels as before) — so $\hat{H}_{\text{dis}}$ shifts each molecule's excited-sublevel energies by its own random offset ϵ_n . Consequences quantified in the paper:

- Dark states gain photon weight $\sum_{\text{dark}} |\langle d | 1_{\text{ph}} \rangle|^2 \approx W^2 / 2g_c^2 \Rightarrow$ they **appear in absorption**.
- Polaritons stay protected while g beats disorder: crossover at $W \sim 2g$.

How Disorder Enters Our Model

We add **Gaussian energy disorder** to the two JT molecules' transition energies:

$$\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2), \quad i = 1, 2, \quad \epsilon = 7.0 \text{ eV}.$$

Per disorder *realization* we build $\hat{H}$ in the ($n_{\text{ex}} = 1, j = -1$) sector, diagonalize, take the intensity of each eigenstate as $|\langle i_0 | \psi \rangle|^2$, broaden with a Lorentzian (FWHM, $\gamma = 0.025 \text{ eV}$), and **average over realizations**.

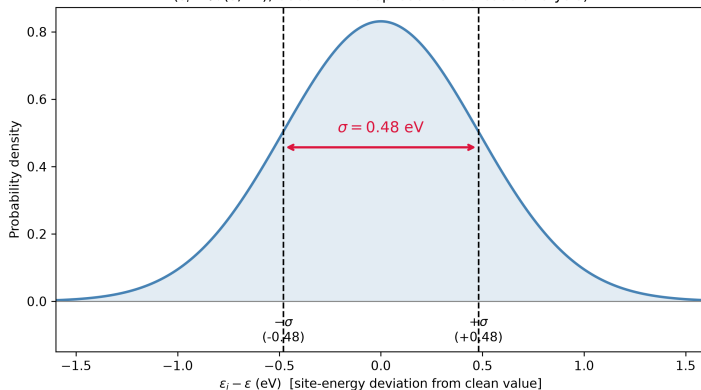
Energy scales

vibrational quantum ω	0.082 eV	JT coupling $\kappa/\omega = 2.2$
per-molecule coupling $g = \Omega/2\sqrt{N}$	0.242 eV	cavity $\omega_c = 6.85 \text{ eV}$
Rabi splitting $\Omega = 2\sqrt{N}g$	0.685 eV	molecular excitation energies $\epsilon = 7.0 \text{ eV}, N = 2$

Disorder strength quoted relative to g : we run at $\sigma \approx 2g$ (strong-disorder band), where the representative cases are physically distinct.

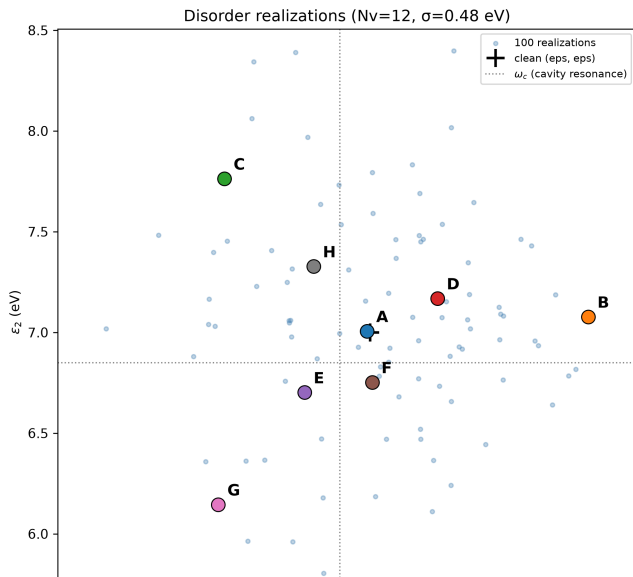
Disorder Realizations: The Gaussian Distribution

Gaussian disorder distribution, $\sigma = 0.48$ eV
($\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2)$, used in the representative-case analysis)



- Each molecule's excitation energy is drawn from $\mathcal{N}(\epsilon, \sigma^2)$ with $\sigma = 0.48$ eV.
- Dashed lines / arrow mark $\pm\sigma$ (-0.48 to $+0.48$ eV).
- We average absorption over 100 such realizations.

Representative-Realization Analysis: The (ϵ_1, ϵ_2) cloud

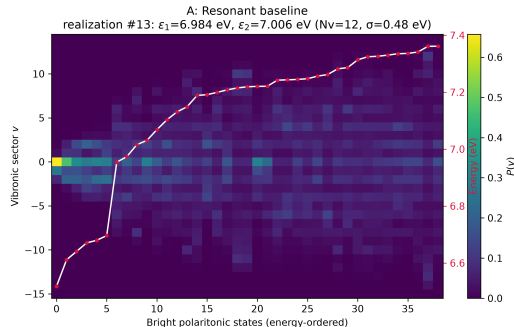
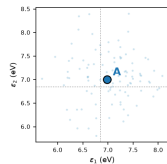
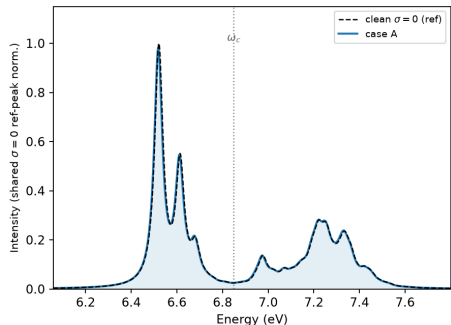


Behind the disorder average lie many individual (ϵ_1, ϵ_2) draws. We pick **9 physically meaningful realizations** (Cases A–I), each best satisfying a criterion on the actual drawn energies:

- average shift $s = (\delta_1 + \delta_2)/2$;
- inter-molecular mismatch $\Delta = \epsilon_1 - \epsilon_2$.

Dotted lines mark ω_c ; the “+” marks the clean (ϵ, ϵ) . Each case is then recomputed individually.

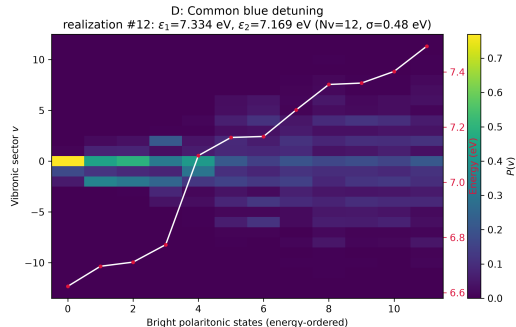
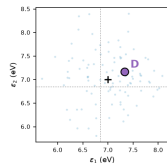
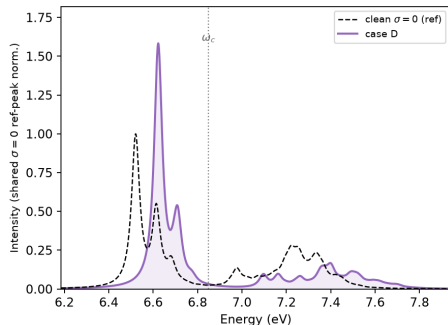
Case A — resonant baseline (no disorder)



Both molecules essentially on resonance, $\epsilon_1 \approx \epsilon_2 \approx \epsilon$.

$(\epsilon_1, \epsilon_2) = (6.98, 7.01)$, $\Delta = 0.02$, $s \approx 0$; **39 bright**, $\text{PR}_{\text{max}} = 20.6$ at $E = 7.361$ eV — reproduces the clean collective cascade.

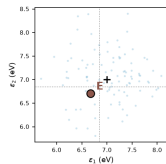
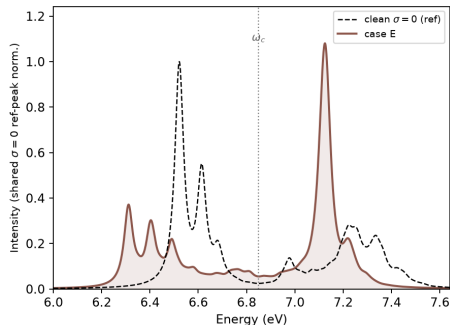
Case D — common blue detuning



From A, shift *both* molecules *up* together by $\sim +g$ (Δ small): a common blue shift.

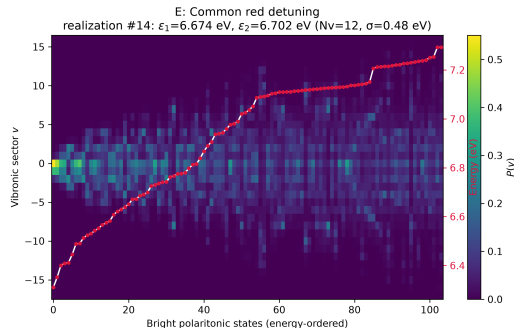
$(\epsilon_1, \epsilon_2) = (7.33, 7.17)$, $\Delta = 0.17$, $s = +1.0g$; **12 bright**, $PR_{\max} = 12.5$ at $E = 7.354$ eV :

Case E — common red detuning

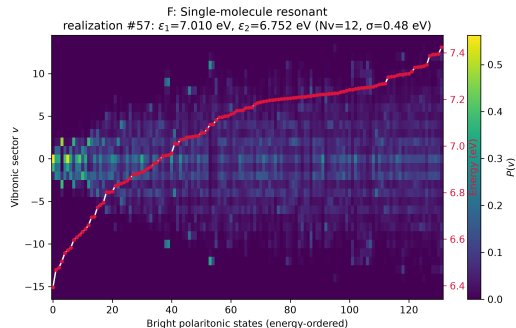
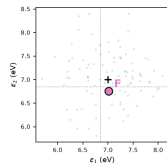
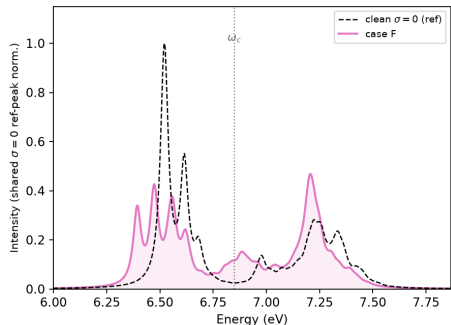


Flip the common shift to $-g$ (mirror of D, but red)

$(\epsilon_1, \epsilon_2) = (6.67, 6.70)$, $\Delta = 0.03$, $s = -1.3g$; **104 bright**, $\text{PR}_{\text{max}} = 22.4$ at $E = 7.215$ eV



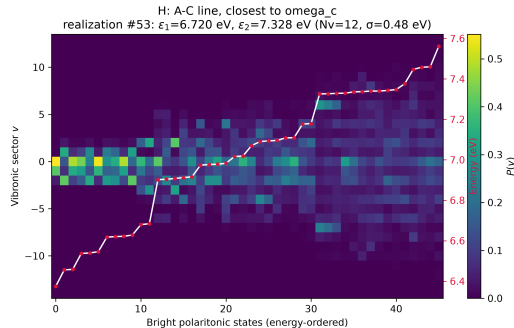
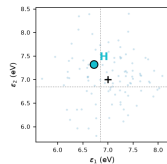
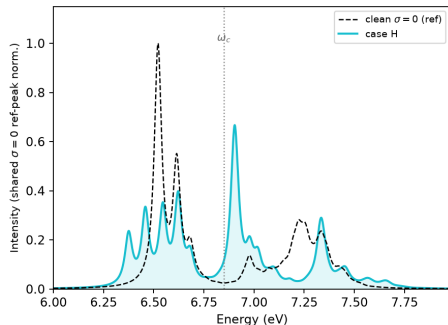
Case F — single-molecule resonant



Now introduce a real inter-molecular mismatch: bring one molecule back *onto* resonance while the other stays detuned by $\gtrsim g$ - a mildly asymmetric *collective* case (not localized).

$(\epsilon_1, \epsilon_2) = (7.01, 6.75)$, $\Delta = g$, $s = -0.5g$; **132 bright**, $\text{PR}_{\text{max}} = 21.6$ at $E = 7.424$ eV.

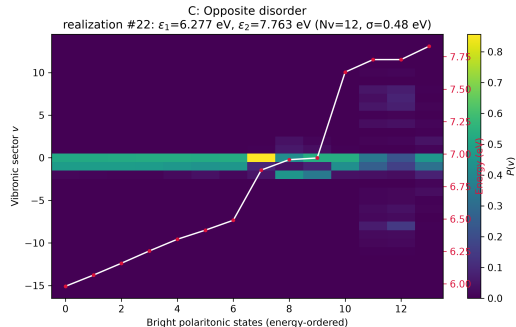
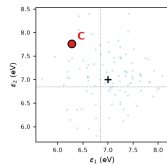
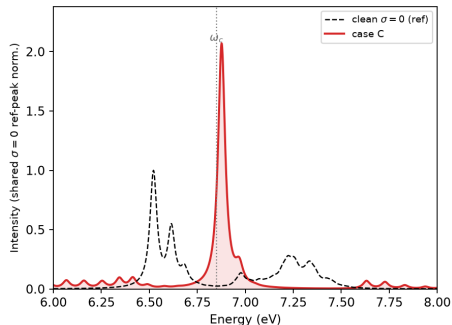
Case H — A-C line, interior (opposite disorder)



Detune the two molecules *oppositely* (one red, one blue) about resonance, $s \approx 0$ — the interior of the line joining A and C.

$(\epsilon_1, \epsilon_2) = (6.72, 7.33)$, $\Delta = 2.5g$, $s \approx 0$; **46 bright**, $\text{PR}_{\text{max}} = 16.2$ at $E = 7.337$ eV — **intermediate**: brightens more peaks than A yet spreads each less.

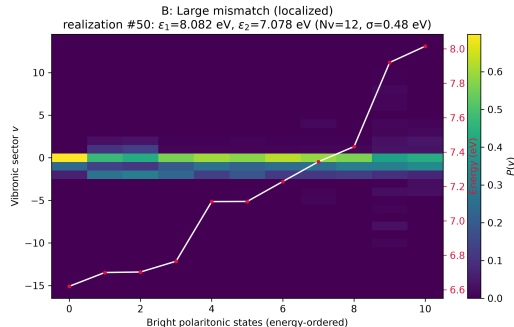
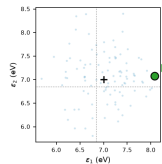
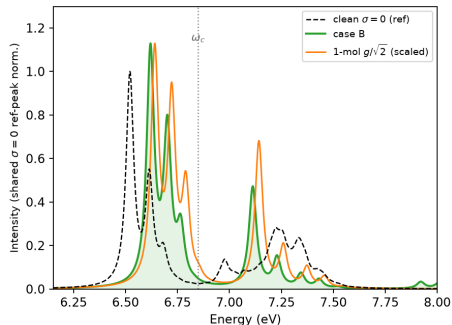
Case C — opposite disorder (far)



Push the opposite detuning *further out* along the same A–C line ($\Delta=6g$): both molecules now leave the bright line.

$(\epsilon_1, \epsilon_2) = (6.28, 7.76)$, $\Delta=6g$, $s \approx 0$; **14 bright**, $\text{PR}_{\text{max}} = 8.5$ at $E = 7.728$ eV

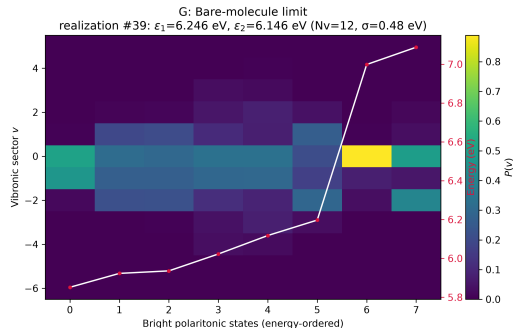
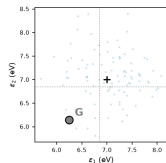
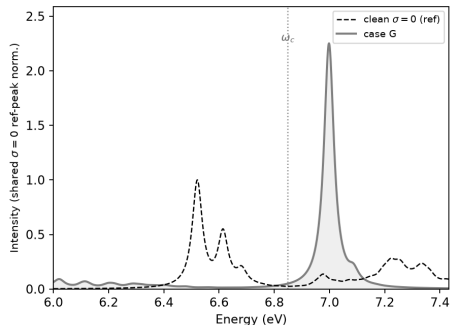
Case B — large same-side mismatch (localized)



Instead of opposite, detune one molecule far on the *same* side ($\Delta=4g$): the excitation *localizes* on the resonant molecule $\Rightarrow$ spectrum $\approx$ a single JT-polariton (orange: 1-mol $g/\sqrt{2}$).

$(\epsilon_1, \epsilon_2) = (8.08, 7.08)$, $\Delta=4g$, $s=+2.4g$; **11 bright**, $PR_{\max} = 3.8$ at $E = 7.921$ eV

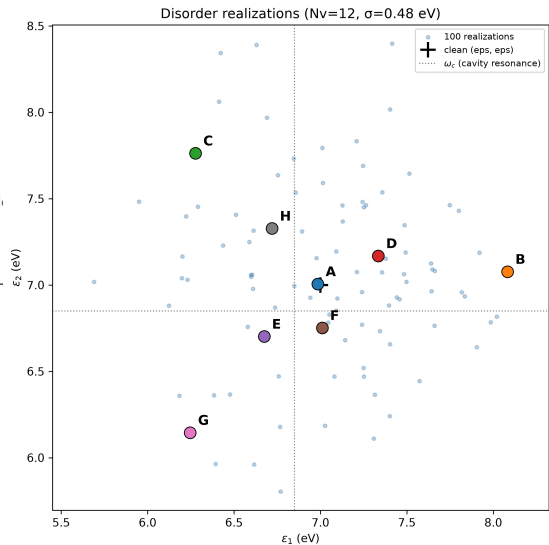
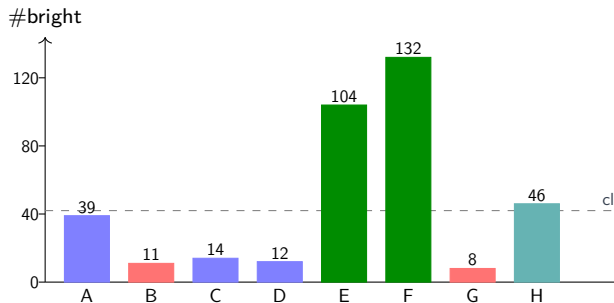
Case G — bare-molecule limit (decoupled)



Finally push *both* molecules far below resonance together : both decouple from the cavity.

$(\epsilon_1, \epsilon_2) = (6.25, 6.15)$, $\Delta = 0.10$, $s = -3.3g$; **8 bright**, $\text{PR}_{\text{max}} = 4.5$ at $E = 6.118$ eV — **dead** (near-bare photon).

Number of Bright Vibronic States vs. Case



Summary

- 1 **Reproduced** the single-molecule cavity-JT polariton spectrum (LP/UP, bounded ν) and the two-molecule collective spectrum + $P(\nu)$ heatmap (unbounded cascade, PR ~ 20).
- 2 **Introduced disorder** following the arrowhead-matrix and disordered Holstein–Tavis–Cummings frameworks: Gaussian excitation-energy disorder $\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2)$ used.
- 3 **Implemented & analyzed** disorder in our $N = 2$ model: realizations, representative cases (A–H), and $P(\nu)$ heatmaps with an LP/UP energy overlay.
- 4 **Key result:** destroys the observed cascade depends on whether it pushes the bright polariton closer to the dense manifold of otherwise-dark states, or pulls it away governed by proximity to cavity resonance and to the collective coupling scale, not disorder magnitude alone.

- ❶ K. R. Nandipati & O. Vendrell, *Cavity Jahn–Teller polaritons in molecules*, Phys. Rev. A **107**, L061101 (2023).
- ❷ S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025).
- ❸ C. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo, D. Hagenmüller, *Large Random Arrowhead Matrices. . .*, arXiv:2105.08444 (2021).
- ❹ D. Wellnitz, G. Pupillo, J. Schachenmayer, *Disorder enhanced vibrational entanglement and dynamics in polaritonic chemistry*, Commun. Phys. **5**, 120 (2022).
- ❺ H. A. Jahn & E. Teller, Proc. R. Soc. A **161**, 220 (1937); (E×e): H. C. Longuet-Higgins et al., Proc. R. Soc. A **244**, 1 (1958).
- ❻ T. W. Ebbesen, *Hybrid light–matter states. . .*, Acc. Chem. Res. **49**, 2403 (2016).